A patient’s history.
A signal’s next chapter.



AURA Twin

A synthetic digital twin for glucose forecasting and clinician-led review.

+120 MINUTE HORIZON

Black Squad

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Give changing signals their historical context.

A focused proof of concept for reviewing possible future high glucose.

01

The history

Static records explain baseline context, but cannot show what changed in the last hour.

02

The stream

Wearable trends show recent change, but do not carry the whole historical profile.

03

The twin

A fused patient state updates a forecast and makes its inputs available for review.

Intended interaction: select a patient → inspect context → review forecast and evidence → document a demo review.

Predict a defined event, at a defined time.

A precise outcome prevents a convincing demo from becoming an ambiguous claim.

EVENT LABEL

Glucose ≥180 mg/dL at exactly t + 120 min

Not “any excursion within the next two hours.”

1 **Numerical trajectory**

Forecasts at +30, +60, +90 and +120 minutes.

2 **Binary probability**

Risk of the exact +120-minute reading meeting the threshold.

3 **Range context**

70–180 mg/dL is a common CGM context; the ≥180 boundary is our explicit convention.

Source: NIDDK, Clinical Targets for Continuous Glucose Monitoring Data. Individual clinical targets can differ.

Wholly synthetic. Fully inspectable.

No real patient records, device accounts or clinical trial data are used.

A

Static / historical EHR

Age, BMI, HbA1c, LDL and diabetes duration.

Generated demographic and clinical context. Arbitrary genotype markers are excluded from prediction.

B

Dynamic / wearable stream

CGM glucose, heart rate, HRV, step count and sleep.

Five-minute readings with synthetic routines, noise and person-level variation.

240 generated patients · 4 simulated days each · fixed seed 20261019

Synthea is a future integration option, not the source of these results. The simulator’s design limits generalization.

The patient state evolves with the stream.

A narrow data-driven twin, not a whole-body physiological simulator.

1 **Stable context**

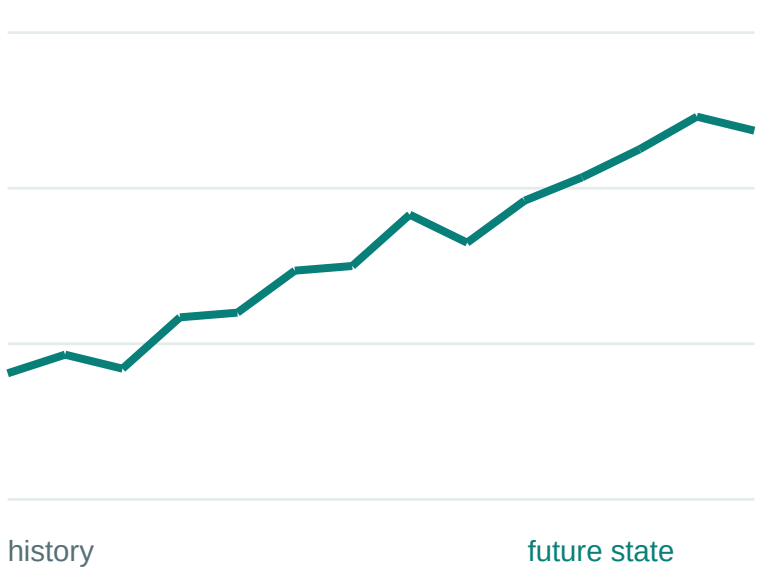
Historical features remain attached to the selected synthetic patient.

2 **Time-local state**

Causal features summarize only observations available at the replay timestamp.

3 **Updated outlook**

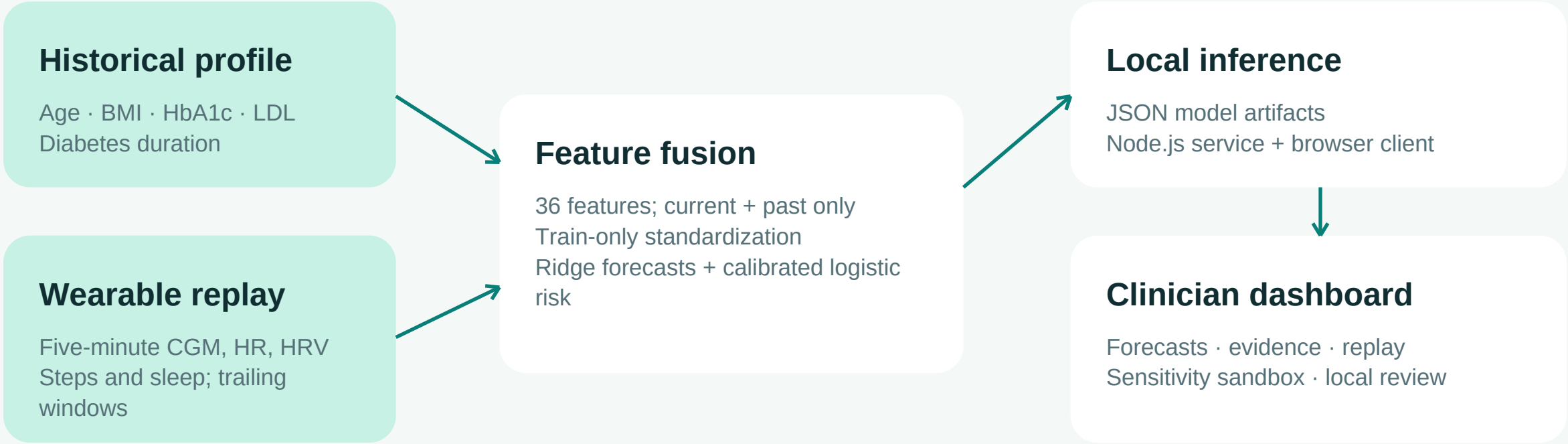
A new prediction follows each replay step; forecast, risk and drivers are inspectable.



Illustrative state trajectory

Two streams. One inspectable patient state.

Implemented local prototype: synthetic data generation, trained models and clinician review.



Future observations are training labels only. They never enter the inference feature window.

Small enough to inspect. Real enough to test.

An exported model powers inference; dashboard values are not fixed mockup numbers.

01

Feature fusion

36 standardized features: EHR context, CGM lags/trends, HR/HRV, activity, sleep, clock harmonics and explicit interactions.

02

Two outputs

Ridge regression forecasts four horizons. Logistic regression estimates the +120-minute event probability.

03

Validation layer

Platt probability calibration and descriptive residual intervals fitted on validation patients.

Coefficient contributions describe model behavior. They do not establish causality.

Split people before fitting models.

Repeated windows from the same synthetic patient never cross train / validation / test partitions.

144

Train

20,160 windows
Fit scaler + model

48

Validation

6,720 windows
Calibration + intervals

48

Test

6,720 windows
Final evidence only

30-minute window stride · 24-hour history · exact future labels · 8.04% test event prevalence

Quantify the prediction and its tradeoffs.

48 held-out synthetic patients, 6,720 correlated windows; alert threshold = 0.20.

10.95

MAE / mg/dL

+120-minute forecast

0.969

AUROC

Event discrimination

0.723

Average precision

8.04% event prevalence

83.9%

Recall at 0.20

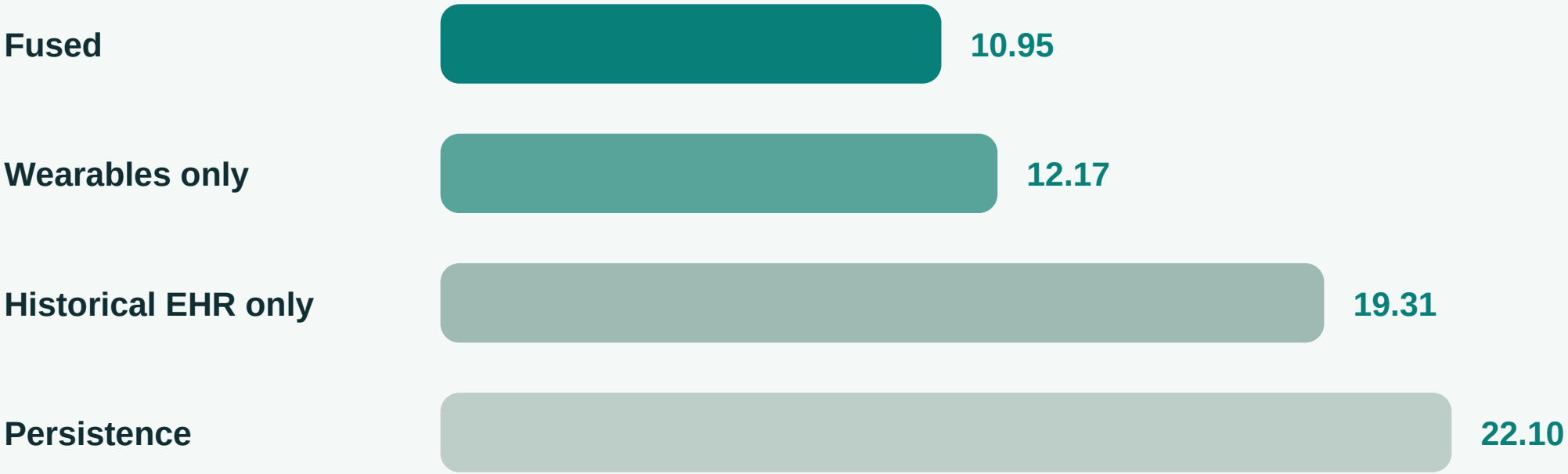
87 / 540 positive windows missed

Brier 0.0372 · Alert precision 53.2% · MAE patient-bootstrap 95% CI 10.29–11.59 mg/dL

These results test a designed simulator. They do not establish clinical safety, effectiveness or real-world accuracy.

Both streams improve this synthetic forecast.

+120-minute mean absolute error; lower is better. No clinical generalization claim.

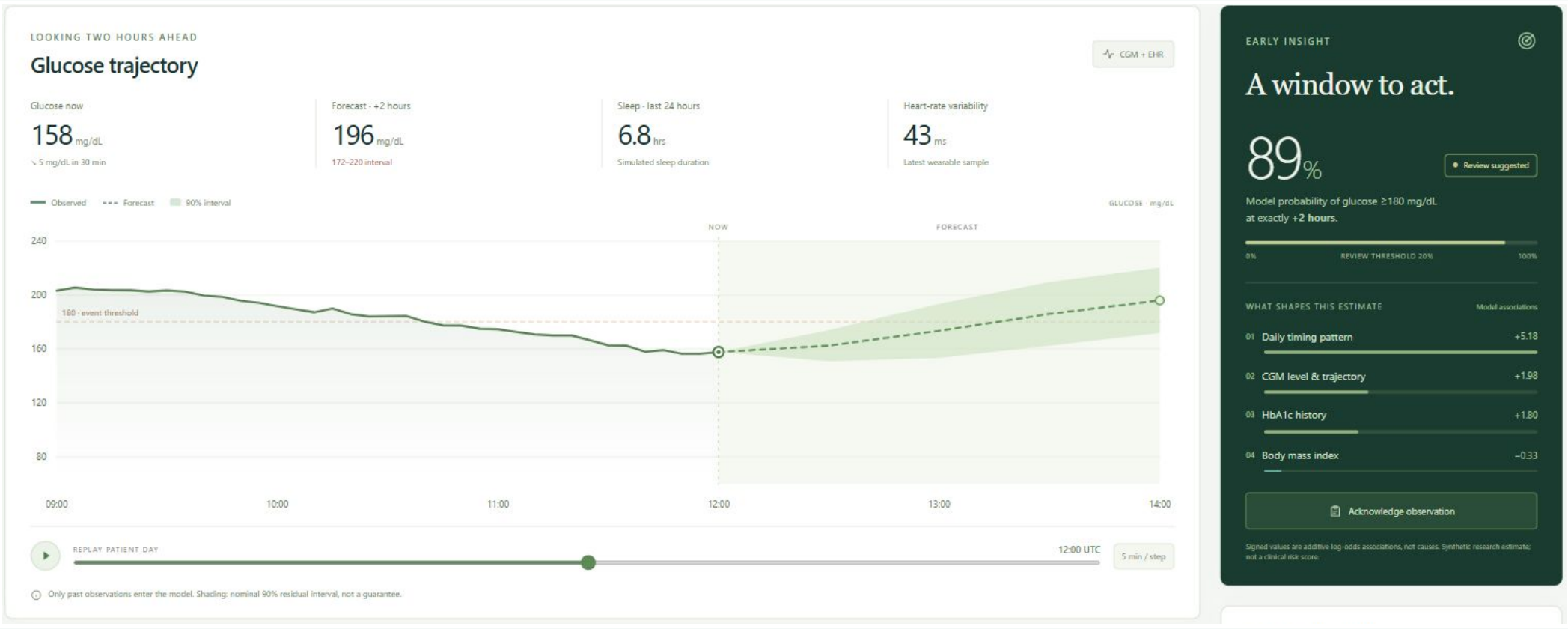


Fusion reduces MAE by 10.1% versus wearables only in this experiment.

Regular routines and HbA1c-linked simulator kinetics can advantage fusion. Persistence copies the current reading.

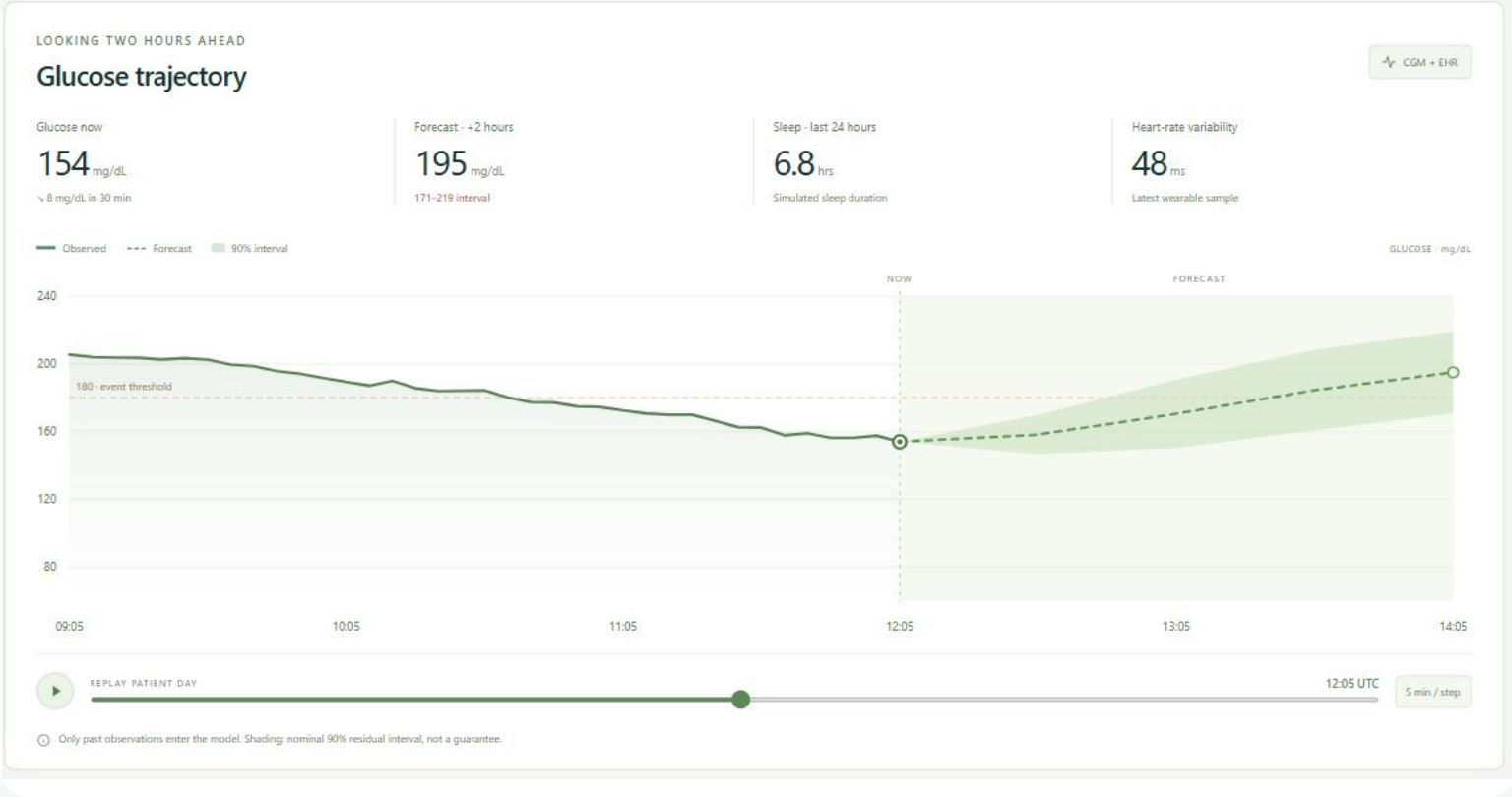
An overview designed for a clinical review.

Patient context, a temporal forecast and supporting evidence share one workspace.



Watch the forecast change as time advances.

Pause, inspect a time point and advance the synthetic stream.



1

Historical versus projected

The future curve is a forecast, not a sensor measurement.

2

Descriptive interval

Validation residuals give a marginal error band.

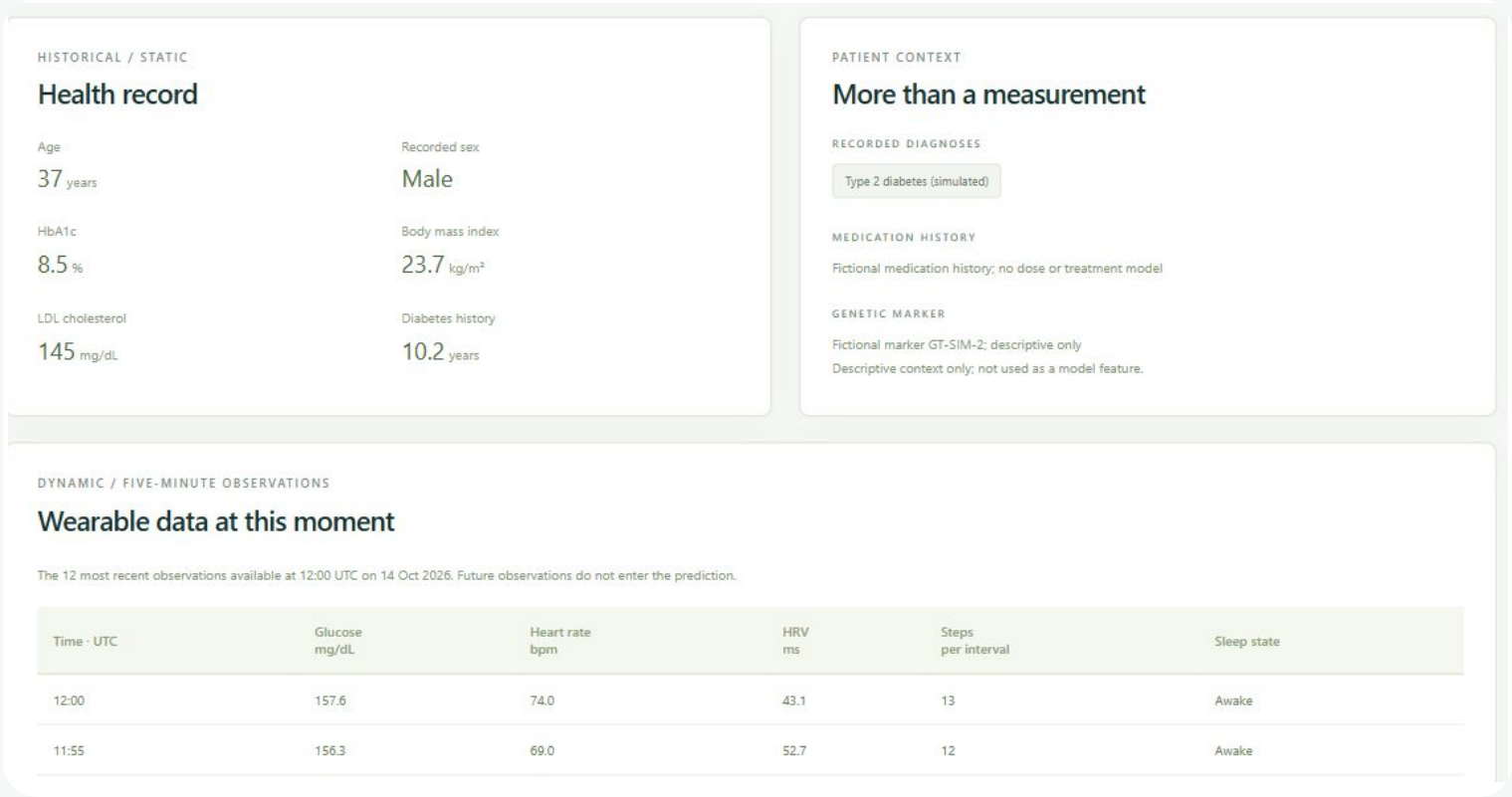
3

Know the limitation

Test coverage: 90.0%; mean width: 48.5 mg/dL. No per-patient guarantee.

Make the model’s inputs visible.

A useful interface lets a reviewer inspect the basis of an estimate.



HISTORICAL / STATIC

Health record

Age 37 years	Recorded sex Male
HbA1c 8.5 %	Body mass index 23.7 kg/m²
LDL cholesterol 145 mg/dL	Diabetes history 10.2 years

PATIENT CONTEXT

More than a measurement

RECORDED DIAGNOSES

Type 2 diabetes (simulated)

MEDICATION HISTORY

Fictional medication history; no dose or treatment model

GENETIC MARKER

Fictional marker GT-SIM-2; descriptive only

Descriptive context only; not used as a model feature.

DYNAMIC / FIVE-MINUTE OBSERVATIONS

Wearable data at this moment

The 12 most recent observations available at 12:00 UTC on 14 Oct 2026. Future observations do not enter the prediction.

Time · UTC	Glucose mg/dL	Heart rate bpm	HRV ms	Steps per interval	Sleep state
12:00	157.6	74.0	43.1	13	Awake
11:55	156.3	69.0	52.7	12	Awake

1

Explanations are mathematical

Feature contributions describe fitted log-odds behavior.

2

They are not causal

Correlated inputs and interactions can complicate interpretation.

3

Keep units explicit

Probability, glucose and log-odds contributions are different quantities.

Explore an assumption. Do not infer a treatment.

Scenario controls are an illustrative arithmetic sandbox, separate from validated causal inference.

01

Baseline prediction

The trained probability remains the baseline model output.

Reset restores the original forecast.

02

Hypothetical scenario

Meal carbohydrate and activity controls modify a displayed forecast through simplified assumptions.

No claim of intervention efficacy.

No medication guidance, dosing, treatment recommendation or causal benefit estimate is produced.

Close the loop with an inspectable demo record.

Local actions demonstrate an interaction pattern; they are not a production clinical record.

01

Select

Search the synthetic patient cohort

→

02

Inspect

Context, trends, and signal quality

→

03

Review

Forecast, probability and model drivers

→

04

Document

Acknowledge alert and save a local note

→

05

Export

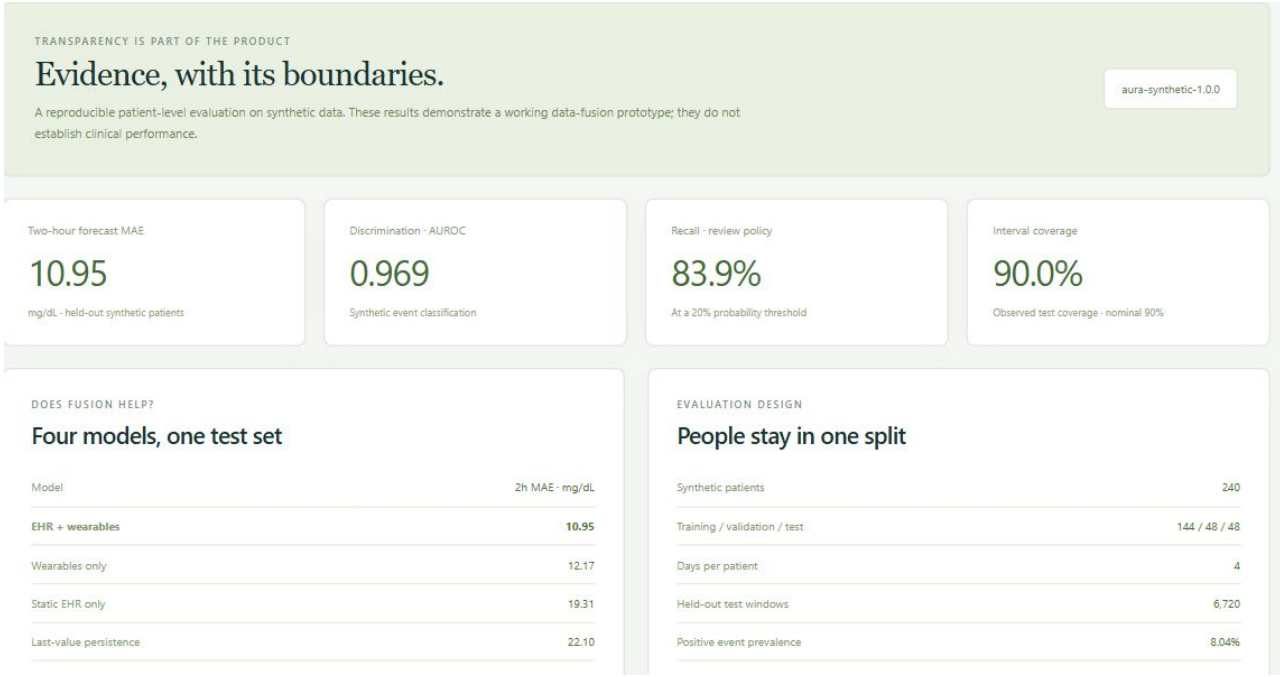
Carry the demo review into a readable file

Acknowledge a demo alert · save local notes · download a clinician summary

Acknowledgement records review, not safety. Authentication, governed audit logs and shared clinical records are future work.

The evidence travels with the code.

The repository contains the generator, training pipeline and exported artifacts.



1 **Generate & train**

Fixed seed and train-only preprocessing.

2 **Inspect & verify**

Readable model, metrics and ablations.

3 **Run & review**

Local Node.js and JavaScript application.

Model version: aura-synthetic-1.0.0

Numerical results are bound to the submitted code, seed and dependencies. Use the root README for exact commands.

The strongest claim is an inspectable prototype.

Clinical benefit, generalization and compliance have not been demonstrated.

Simulator bias

Designed routines and HbA1c-linked kinetics may make prediction easier.

Alert tradeoff

At threshold 0.20: 87 false-negative and 398 false-positive test windows.

Distribution shift

New populations, devices, medications and unusual routines are untested.

Governance gap

No clinical authentication, consent workflow, regulated validation or production audit trail.

Synthetic development is not proof of DPDP / HIPAA compliance or medical-device readiness.

Validate in stages before pursuing deployment.

Progress should be gated by evidence, not the sophistication of the interface.

01

Stress tests

Missingness, clock shifts, routine changes, subgroup checks.

02

Independent data

Appropriate permissions; external validation and recalibration.

03

Silent study

Prospective performance, alert burden and missed-event analysis.

04

Clinical workflow

Usability, escalation ownership, security and governance.

Real device / EHR connections and any clinical deployment remain future work.

A complete local package. Clear publishing steps.

Black Squad · Amrita Vishwa Vidyapeetham, Chennai

01

Included locally

Working application and source
Synthetic data and model artifacts
Training and evaluation code
Architecture + presentation PDFs
MIT license and technical documentation

02

Final team handoff

Verify the real demonstration video
Publish a public GitHub repository
Insert accessible links into README
Test every link while signed out
Enter leader details in the portal

Leader: Banu Sreya V. · Member: Duvvuru Deepak Reddy · Public contact details intentionally omitted.

From two streams to one reviewable outlook.

A working fusion pipeline.
A measurable two-hour target.
A clinician-oriented interface.
Evidence with explicit boundaries.

AURA TWIN

Synthetic prototype · no clinical validation or treatment recommendation